Polynomial central limit theorems for the canonical Gaussian measure of a weighted backward shift

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Lane 14, candidate partial result

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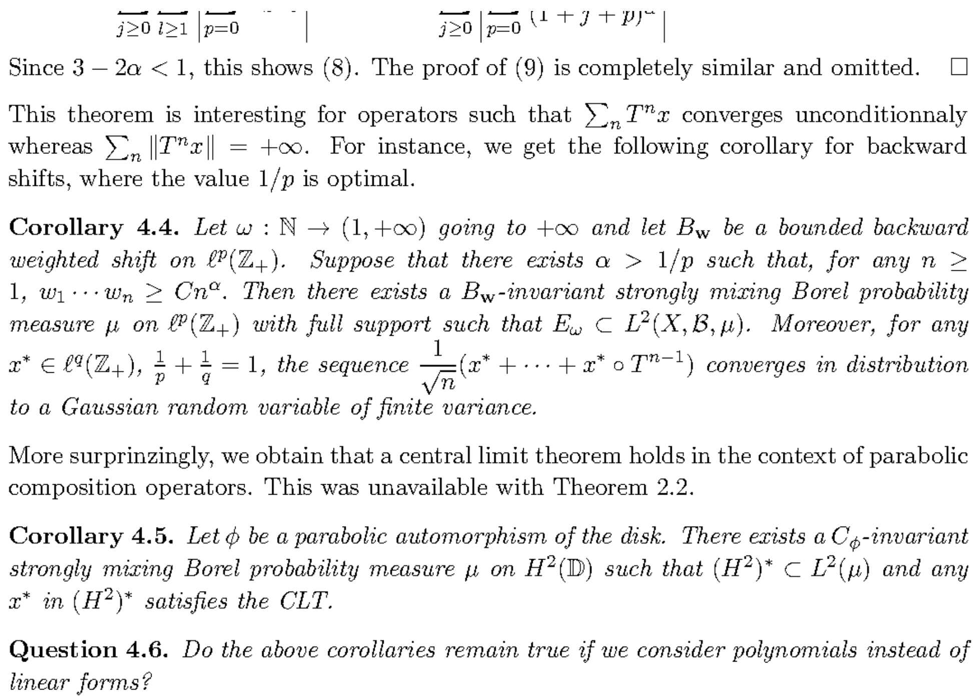
Status. Candidate substantial partial result, likely valid pending expert review. The theorem below completely answers the weighted-backward-shift half of Bayart's Question 4.6 and strengthens its hypothesis. The parabolic-composition-operator half is not claimed.

1 Source question and result

The source is Frédéric Bayart, *Central limit theorems in linear dynamics*, arXiv:1304.2621, published in *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques* 51 (2015), 1131–1158 [1]. On page 30, after a weighted-shift corollary and a parabolic-composition-operator corollary, it asks:

> **Question 4.6.** Do the above corollaries remain true if we consider polynomials instead of linear forms?



$\sum_{j\geq 0}\sum_{l\geq 1}\Big|\sum_{p=0}\Big| \qquad \sum_{j\geq 0}\Big|\sum_{p=0}(1+j+p)\Big|$

Since $3-2\alpha<1$, this shows (8). The proof of (9) is completely similar and omitted. □

This theorem is interesting for operators such that $\sum_n T^n x$ converges unconditionnaly whereas $\sum_n \|T^n x\| = +\infty$. For instance, we get the following corollary for backward shifts, where the value $1/p$ is optimal.

Corollary 4.4. *Let $\omega : \mathbb{N} \to (1,+\infty)$ going to $+\infty$ and let $B_{\mathbf{w}}$ be a bounded backward weighted shift on $\ell^p(\mathbb{Z}_+)$. Suppose that there exists $\alpha > 1/p$ such that, for any $n \geq 1$, $w_1 \cdots w_n \geq Cn^\alpha$. Then there exists a $B_{\mathbf{w}}$-invariant strongly mixing Borel probability measure μ on $\ell^p(\mathbb{Z}_+)$ with full support such that $E_\omega \subset L^2(X,\mathcal{B},\mu)$. Moreover, for any $x^* \in \ell^q(\mathbb{Z}_+)$, $\frac{1}{p}+\frac{1}{q}=1$, the sequence $\frac{1}{\sqrt{n}}(x^* + \cdots + x^* \circ T^{n-1})$ converges in distribution to a Gaussian random variable of finite variance.*

More surprinzingly, we obtain that a central limit theorem holds in the context of parabolic composition operators. This was unavailable with Theorem 2.2.

Corollary 4.5. *Let ϕ be a parabolic automorphism of the disk. There exists a C_ϕ-invariant strongly mixing Borel probability measure μ on $H^2(\mathbb{D})$ such that $(H^2)^* \subset L^2(\mu)$ and any x^* in $(H^2)^*$ satisfies the CLT.*

Question 4.6. *Do the above corollaries remain true if we consider polynomials instead of linear forms?*

Figure 1: Corollaries 4.4–4.5 and Question 4.6 in the source PDF, page 30.

For the weighted-shift half, the answer is affirmative under the exact summability condition that already characterizes the existence of an invariant full-support measure. In particular, the pointwise polynomial growth condition in the source corollary is unnecessary.

Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ and $1 \le p < \infty$. For a bounded sequence of positive weights $\mathbf{w} = (w_n)_{n\ge1}$, write

$$B_{\mathbf{w}}(x_0, x_1, \ldots) = (w_1x_1, w_2x_2, \ldots), \qquad W_0 = 1, \quad W_n = w_1 \cdots w_n,$$

and set $s_n = W_n^{-1}$. A polynomial below is a continuous real-valued polynomial on the real Banach space underlying $\ell^p(\mathbb{N}_0;\mathbb{K})$, exactly as in the source.

Theorem 1 (Polynomial CLT for the canonical Gaussian shift)**.** *Suppose*

$$\sum_{n\ge0} W_n^{-p} < \infty. \tag{1}$$

Let $(\gamma_n)_{n\ge0}$ be independent standard real Gaussians when $\mathbb{K} = \mathbb{R}$, and independent standard circular complex Gaussians when $\mathbb{K} = \mathbb{C}$. Let μ be the law on $\ell^p(\mathbb{N}_0;\mathbb{K})$ of

$$X = (s_n\gamma_n)_{n\ge0}.$$

Then μ is a $B_{\mathbf{w}}$-invariant strongly mixing Gaussian Borel probability measure with full support. Every continuous real polynomial P belongs to $L^q(\mu)$ for all $q < \infty$. Moreover, with

$$\overline{P} = P - \int P \, d\mu,$$

there is a number

$$\sigma_P^2 = \sum_{k\in\mathbb{Z}} \operatorname{Cov}\big(\overline{P}(X), \overline{P}(B_{\mathbf{w}}^k X)\big) \ge 0,$$

where the negative indices are interpreted in the stationary natural extension and the series is absolutely convergent, such that

$$\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \overline{P} \circ B_{\mathbf{w}}^k \xrightarrow[N\to\infty]{\text{law}} \mathcal{N}(0, \sigma_P^2).$$

The Gaussian law is allowed to be degenerate.

Remark 1 (Relation to Corollary 4.4)**.** *If $W_n \ge Cn^\alpha$ for some $\alpha > 1/p$, then $\sum_n W_n^{-p} \le C^{-p} \sum_n n^{-\alpha p} < \infty$. Thus Theorem 1 answers the weighted-shift part of Question 4.6 under every hypothesis of the source corollary, and in fact under the strictly weaker condition (1).*

2 Proof intuition

The canonical Gaussian measure turns the orbit into a nonlinear Bernoulli shift:

$$P(B_{\mathbf{w}}^k X) = P\big((s_n\gamma_{n+k})_{n\ge0}\big).$$

Replacing the innovation γ_0 changes exactly one coordinate in each affected time slice. A global Lipschitz estimate would discard this geometry and demand the false extra hypothesis $s \in \ell^1$. Instead, expand P along the changed coordinate. The term of Taylor order r contains s_n^r multiplied by the diagonal coefficient $D^rP(x)[e_n, \ldots, e_n]$.

The decisive deterministic fact is that the diagonal of an r-linear form on ℓ^p lies in $\ell^{p/(p-r)}$ when $r < p$. This exponent is exactly Hölder-conjugate to p/r, the exponent for (s_n^r). The derivative coefficients are random Gaussian polynomials, but fixed-degree Gaussian norm equivalence makes the deterministic diagonal estimate survive in L^2. Hence the physical-dependence coefficients are summable. The central limit theorem of El Machkouri–Volný–Wu [2] then finishes the proof.

3 Three auxiliary facts

Lemma 1 (Block-diagonal coefficients on ℓ^p)**.** *Regard $\ell^p(\mathbb{N}_0;\mathbb{K})$ as a real Banach space. Let $A : (\ell^p(\mathbb{N}_0;\mathbb{K}))^r \to \mathbb{R}$ be a continuous r-linear form. For every $j \in \{1,\dots,r\}$, let $u_n^{(j)}$ be supported in the n-th scalar coordinate and satisfy $\left\|u_n^{(j)}\right\| \le 1$, and put $a_n = A(u_n^{(1)},\dots,u_n^{(r)})$. If $r < p$ and $t = p/(p-r)$, then*

$$\|(a_n)\|_{\ell^t} \le C_r \|A\| .$$

If $r \ge p$, then $\sup_n |a_n| \le \|A\|$.

Proof. The second statement is immediate. For the first, complexify the real Banach space and A; this changes the norm by a constant depending only on r. For finitely supported $y = (y_n)$ choose b_n with $b_n^r = y_n$. Let (ζ_n) be independent variables uniformly distributed over the r-th roots of unity (for $r = 1$, use ordinary duality). Since a product of r such roots has nonzero expectation only when all r indices coincide,

$$\sum_n a_n y_n = \mathbb{E}_\zeta A\left(\sum_n b_n\zeta_n u_n^{(1)},\dots,\sum_n b_n\zeta_n u_n^{(r)}\right).$$

Therefore

$$\left|\sum_n a_n y_n\right| \le C_r \|A\| \left(\sum_n |b_n|^p\right)^{r/p} = C_r \|A\| \|y\|_{\ell^{p/r}} .$$

Duality between $\ell^{p/r}$ and $\ell^{p/(p-r)}$ proves the claim. $\square$

Lemma 2 (Uniform Gaussian polynomial estimates)**.** *Fix an integer $m \ge 0$ and $1 \le q < \infty$. There is $C_{m,q}$ such that every real polynomial Q of degree at most m in an arbitrary finite or countable Gaussian family satisfies*

$$\|Q\|_2 \le C_{m,q} \|Q\|_q . \tag{2}$$

Moreover, if G is one distinguished standard Gaussian and $Q(0,\widehat{G})$ denotes evaluation at $G = 0$, then

$$\left\|Q(0,\widehat{G})\right\|_2 \le C_m \left\|Q(G,\widehat{G})\right\|_2 . \tag{3}$$

The constants do not depend on the number of Gaussian variables.

Proof. For $q \ge 2$, (2) is immediate. For $1 \le q < 2$, Gaussian hypercontractivity gives $\|Q\|_4 \le 3^{m/2}\|Q\|_2$. Applying Paley–Zygmund to Q^2 shows that $|Q|$ exceeds a fixed multiple of $\|Q\|_2$ on a set of probability bounded below by a constant depending only on m. This implies (2).

For (3), expand in Hermite polynomials of G:

$$Q(G,\widehat{G}) = \sum_{j=0}^{m} c_j(\widehat{G})H_j(G).$$

Orthogonality gives $\|Q\|_2^2 = \sum_{j=0}^m j!\,\|c_j\|_2^2$, while $Q(0,\widehat{G}) = \sum_{j=0}^m H_j(0)c_j(\widehat{G})$. Cauchy–Schwarz over this finite sum proves the uniform bound. Countably many variables follow by the usual L^2 approximation. $\square$

Lemma 3 (Summable physical dependence)**.** *Let $P : \ell^p(\mathbb{N}_0;\mathbb{K}) \to \mathbb{R}$ be a continuous polynomial and assume $s = (s_n) \in \ell^p$. On a two-sided iid standard Gaussian sequence $(\gamma_j)_{j\in\mathbb{Z}}$ over $\mathbb{K}$ define*

$$Y_k = P\big((s_n\gamma_{n+k})_{n\ge0}\big) - \mathbb{E}P\big((s_n\gamma_n)_{n\ge0}\big).$$

Let Y_k^* be obtained by replacing only γ_0 by an independent copy and set $\delta_{k,2} = \|Y_k - Y_k^*\|_2$. Then

$$\Delta_2 := \sum_{k\in\mathbb{Z}} \delta_{k,2} < \infty.$$

Proof. Write $d = \deg P$. If $k > 0$, then Y_k does not use γ_0, so $\delta_{k,2} = 0$. Fix $n \geq 0$ and consider $k = -n$. Replacing γ_0 changes only coordinate n. Let $\widehat{X}_n$ have the law of X with its n-th coordinate set to zero, and let g, g' be independent standard Gaussians, independent of $\widehat{X}_n$. In the real case, the exact finite Taylor expansion from the common base point $\widehat{X}_n$ gives

$$P(\widehat{X}_n + s_n g e_n) - P(\widehat{X}_n + s_n g' e_n) = \sum_{r=1}^{d} \frac{s_n^r (g^r - (g')^r)}{r!} D^r P(\widehat{X}_n)[e_n, \ldots, e_n].$$

Independence and the triangle inequality imply

$$\delta_{-n,2} \leq \sum_{r=1}^{d} C_r s_n^r b_{n,r}, \qquad b_{n,r} := \left\| D^r P(\widehat{X}_n)[e_n, \ldots, e_n] \right\|_2. \tag{4}$$

In the complex case, write a circular Gaussian in its two independent real components and expand the same real Fréchet derivative. At order r this produces at most 2^r terms, with e_n in each slot replaced by either e_n or ie_n. The proof below applies term by term using the block-diagonal form of Lemma 1; all constants still depend only on d and p. Thus it is enough to continue with the displayed real notation.

Set $Q_{n,r} = D^r P(X)[e_n, \ldots, e_n]$ in the real notation (and use each of the finitely many mixed e_n, ie_n coefficients in the complex case). This is a Gaussian polynomial of degree at most $d - r$. Lemma 2, applied to the distinguished coordinate γ_n (or its two real components), yields

$$b_{n,r} \leq C_d \left\| Q_{n,r} \right\|_2. \tag{5}$$

Suppose first that $r < p$ and put $t = p/(p - r)$. The norm equivalence in Lemma 2, followed by Lemma 1 pointwise, gives

$$\begin{aligned} \left(\sum_n \|Q_{n,r}\|_2^t \right)^{1/t} &\leq C_{d,t} \left(\sum_n \|Q_{n,r}\|_t^t \right)^{1/t} \\ &= C_{d,t} \left(\mathbb{E} \sum_n |Q_{n,r}|^t \right)^{1/t} \\ &\leq C_{d,p,r} \left(\mathbb{E} \|D^r P(X)\|^t \right)^{1/t} < \infty. \end{aligned}$$

The last moment is finite because $D^r P$ has polynomial growth and Gaussian measures have moments of every order. Thus $(b_{n,r}) \in \ell^{p/(p-r)}$. Also $(s_n^r) \in \ell^{p/r}$, so Hölder's inequality gives $\sum_n s_n^r b_{n,r} < \infty$.

If $r \geq p$, then

$$\sup_n \|Q_{n,r}\|_2 \leq \| \|D^r P(X)\| \|_2 < \infty,$$

and (5) makes $(b_{n,r})$ bounded. Since $s \in \ell^p$ and $r \geq p$, one has $\sum_n s_n^r < \infty$. Summing (4) over n completes the proof. □

4 Proof of the main theorem

Proof of Theorem 1. Since

$$\mathbb{E} \sum_{n \geq 0} |s_n \gamma_n|^p = \mathbb{E}|\gamma_0|^p \sum_{n \geq 0} s_n^p < \infty,$$

X belongs to $\ell^p(\mathbb{N}_0; \mathbb{K})$ almost surely. Its law is a Radon Gaussian measure. It has full support: finite coordinate boxes have positive probability, and the Gaussian tail in ℓ^p can be made small with positive probability. The identity $w_{n+1} s_{n+1} = s_n$ gives

$$B_{\mathbf{w}} X = (s_n \gamma_{n+1})_{n \geq 0} \overset{\text{law}}{=} X,$$

so μ is invariant. Cylinder functions of X and of $B_{\mathbf{w}}^{;k}X$ depend on disjoint Gaussian coordinate blocks for all sufficiently large k. Approximation of bounded measurable functions by cylinders proves strong mixing.

A continuous polynomial has polynomial growth. Fernique's theorem therefore implies $P \in L^q(\mu)$ for all finite q.

The process $(Y_k)_{k\in\mathbb{Z}}$ from Lemma 3 is a stationary Bernoulli random field of the form treated by El Machkouri–Volný–Wu [2, Definitions 1–2], and its nonnegative-time restriction has the same law as $(\overline{P}(B_{\mathbf{w}}^{;k}X))_{k\geq 0}$. Lemma 3 gives $\Delta_2 < \infty$. Proposition 2 of [2] yields absolute covariance summability and

$$\frac{1}{N}\mathbb{E}\left(\sum_{k=0}^{N-1} Y_k\right)^2 \longrightarrow \sigma_P^2 = \sum_{k\in\mathbb{Z}} \mathbb{E}(Y_0 Y_k) \geq 0.$$

If $\sigma_P^2 > 0$, the partial-sum variance tends to infinity, so Theorem 1 of [2], with the intervals $\{0, \ldots, N-1\} \subset \mathbb{Z}$, gives convergence to $\mathcal{N}(0, \sigma_P^2)$. If $\sigma_P^2 = 0$, the displayed variance convergence says directly that $N^{-1/2}\sum_{k=0}^{N-1} Y_k \to 0$ in L^2. This is the claimed degenerate Gaussian limit. □

5 Limitations, upgrade attempts, and novelty status

The proof settles the weighted-shift half of Question 4.6 but not the parabolic-composition half. Two focused extensions were tried. The natural Gaussian measure associated with the parabolic eigenvector field has only the source's Hölder exponent $< 1/2$, which does not yield summable dependence. The source's alternative Bernoulli coding has a summable aggregate estimate for linear forms, but individual translated coding vectors decay only like $n^{-\beta}$ with $\beta < 1$. A polynomial Taylor expansion replaces the linear cancellation by random derivative coefficients, and the argument no longer gives a summable physical-dependence series. This is the present obstruction, not a claimed impossibility theorem.

The proof was also tested against two internal hazards. First, a global Lipschitz estimate would impose $s \in \ell^1$ and is not sharp. Second, Taylor expansion around a vector containing the replacement Gaussian would make the scalar increment and derivative coefficient dependent. Expanding from the coordinate-deleted vector and using Lemma 2 repairs that issue.

On 11 August 2026, bounded searches of the run indexes and the web used arXiv id 1304.2621, the exact wording of Question 4.6, the paper title and author, and combinations of "weighted backward shift", "Gaussian invariant measure", "polynomial", and "central limit theorem". They found the source and general CLT references, but no later resolution of Question 4.6 or statement of Theorem 1. Novelty confidence is therefore moderate, pending specialist review and a broader citation search.

Human-review recommendation

The result should receive expert review as a likely valid substantial partial. The highest-value checks are: (i) the real-to-complex reduction in Lemma 1; (ii) the uniform coordinate-deletion estimate in Lemma 2; and (iii) the positive-variance versus zero-variance split in the use of [2]. The parabolic half should remain explicitly outside the claim.

References

[1] F. Bayart, *Central limit theorems in linear dynamics*, Ann. Inst. Henri Poincaré Probab. Stat. **51** (2015), 1131–1158; arXiv:1304.2621, DOI: 10.1214/13-AIHP585.

[2] M. El Machkouri, D. Volný, and W. B. Wu, *A central limit theorem for stationary random fields*, Stochastic Process. Appl. **123** (2013), 1–14; arXiv:1109.0838, DOI: 10.1016/j.spa.2012.08.014.